Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/29/2005
PI Name:	Benedek, George B. Ph.D.		
Project Title:	Kinetic Evolution of Stable and Metastable States in Protein Solutions		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Polymers		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	gbb@mit.edu	**Fax:**	FY 617-225-2585
PI Organization Type:	UNIVERSITY	**Phone:**	617-253-4828
Organization Name:	Massachusetts Institute of Technology		
PI Address 1:	Building 13-2005, Materials Processing Center		
PI Address 2:	77 Massachusetts Avenue		
PI Web Page:	http://web.mit.edu/		
City:	Cambridge	**State:**	MA
Zip Code:	02139-4307	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	05/10/2000	**End Date:**	11/30/2004
No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Gorti, Sridhar	**Contact Phone:**	256 544-0158
Contact Email:	sridhar.gorti@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Lomakin, A. (Massachusetts Institute of Technology) Asherie, N. (Massachusetts Institute of Technology) Annunziata, O. (Massachusetts Institute of Technology)		
Grant/Contract No.:	NAG8-1659		
Performance Goal No.:			
Performance Goal Text:			
	Most investigations of proteins focus primarily on the structure and biological function of individual macromolecules. However, both industrial processes and basic research studies require the production, purification, and stable storage of large quantities of protein, under conditions quite different from that found in vivo. Consequently, it becomes essential to understand the bulk equilibrium and dynamic properties of protein solutions from a biomolecular materials point of view. The scientific approaches involved are the thermodynamics, statistical mechanics, and kinetics of phase transitions. The fundamental operational tool needed for the description and control of biomolecular processing is the phase diagram. The phase diagram of concentrated protein solutions bears a similarity to that of simpler systems. However, in protein solutions numerous metastable phases are possible due to the amphiphilic and polyelectrolyte character of the folded		

Task Description:

polypeptide chains of the proteins. We propose to study experimentally the kinetic evolution of the various metastable states in protein solutions, and determine the degree to which these states control the evolution of the stable equilibrium phases. We shall also establish relationships between the observed kinetics and the interactions between the protein macromolecules. The objective of this research is to study the kinetics of stable and metastable transformations in protein solutions. We will investigate the kinetically controlled competition amongst liquid-liquid phase separation, aggregation, gelation and crystallization. We will also establish a connection between the macroscopic kinetic behavior and the microscopic protein interactions. To do so we propose the following specific aims:

1. Use quasielastic light scattering to study the diffusive transport of proteins in highly concentrated solutions.

2. Measure the temporal evolution of spinodal decomposition and droplet growth in protein solutions exhibiting liquid-liquid phase separation.

3. Determine the generic conditions under which aggregation occurs. Examine the kinetics of this phenomenon and its effect on the formation of condensed protein phases.

4. Develop models using computer simulations and analytic theories to account for the experimentally observed kinetic behavior. The proposed research will provide a sound experimental and theoretical basis for understanding the kinetic evolution of both metastable and stable phases in protein solutions. Indeed, this work should provide important insights into the role of metastable liquid-liquid phase separation, amorphous aggregation and gelation as states which kinetically inhibit the formation of equilibrium phases such as protein crystals or soluble protein solutions. The knowledge gained from the proposed research will have application to basic biological problems such as the self-assembly of supramolecular structures, fibrillogenesis, and the crystallization of proteins for use in x-ray structure determinations. An understanding of the kinetics of phase transformations in protein solutions will also be useful in the optimization of processes such as large-scale separation and purification of recombinant proteins in the biotechnology industry. Furthermore, our investigations can assist in the development of methods to insure the long-term stability of concentrated protein solutions as is required in the pharmaceutical industry.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Introduction
Proteins in solution may be subject to crystallization, liquid-liquid phase transition, self-assembly, aggregation and gelation. Understanding and controlling these transformations is important for the processing and the use of biomaterials, protein crystal growth and protein condensation diseases. We have experimentally and theoretically investigated these phase transition in model proteins. Specifically, we have used the crystallins, a family of eye-lens proteins, as model proteins. We have investigated how changes in the primary structure of these proteins and changes in the solvent composition affect the observed phase behavior.

1. The effect of polyethylene glycol on the liquid-liquid phase transition of protein solutions

We have investigated the effect of polyethylene glycol (PEG) on the liquid-liquid phase transition in protein solution. This investigation was motivated by the extensive practical use of PEG as a precipitating agent in protein crystallization and liquid-liquid phase separation (LLPS). LLPS is important because provides information on protein-protein interactions. This information can be used to optimize the separation and purification of proteins. LLPS is also believed to be relevant for the kinetics of protein crystallization. Finally, LLPS is a powerful tool for the analysis of protein-PEG interactions. In order to understand the action of PEG, we have used bovine gD crystallin as a model system to study the effect of low molecular weight PEG on the protein liquid-liquid phase transition. The aqueous gD crystallin system displays a liquid-liquid coexistence curve with an upper consolute critical temperature of 278K. Measurements of phase separation temperatures as a function of both protein and PEG concentration were determined for several PEG molecular weights. In addition, both PEG and protein concentrations in the coexisting phases were also obtained. The results show that PEG produces an increase of the critical temperature, while leaving the protein critical concentration unchanged. The critical temperature increases with both PEG concentration and molecular weight. We demonstrated that PEG partitioning occurs between the two coexisting phases. We interpreted our experimental results using depletion force theories (macromolecular crowding) better established for colloidal systems. We have also used the topology of the phase diagram to determine the protein-PEG interactions. We found that protein-PEG interactions can only qualitatively be described by depletion forces (i.e. repulsive excluded volume interactions). Indeed, we found that an attractive component, which partially compensates the repulsive excluded volume interactions, must be considered. This investigation indicates that not always PEG can be used as an effective crystallizing agent and that measurements of liquid-liquid phase separation can be used to determine the effectiveness of PEG as a precipitating agent. This work also shows that PEG can be used for inducing liquid-liquid phase transitions of protein aqueous systems.

Ref. Annunziata, O., Asherie, N., Lomakin, A., Pande, J., Ogun, O., and Benedek, G.B. “Effect of polyethylene glycol on the liquid-liquid phase transition of protein solutions,” Proc. Natl. Acad. Sci. USA 99, 14165-14170 (2002).

2. Inducing liquid-liquid phase separation in protein solutions

There are protein-water binary systems for which LLPS is not experimentally detectable. If this transition is simply inaccessible, i.e., the phase separation temperature falls below the freezing point for example, the addition of PEG can be used to reveal LLPS. Our previous work on gD crystallin+PEG+water ternary systems has shown that the addition of PEG increases the LLPS temperature of protein solutions. We have successfully applied this technique to reveal LLPS in two protein-water systems: gS crystallin (gS) and thaumatin.

gS protein-water binary system undergoes neither crystallization nor LLPS for temperatures higher than 263 K. Small amounts of PEG with average molecular weight 1450 g×mol-1 (PEG1450) were added to the protein-water solution and LLPS was observed. Measurements of phase separation temperatures as a function of both protein and PEG concentration were determined for several PEG molecular weights. In addition, both PEG and protein concentrations in the coexisting phases were also obtained. We estimated the critical temperature, , for the binary gS-water system. We obtain: 245 K for the binary gS-water system. The value of the critical temperature, , provides a measure of the magnitude of the protein-protein attractive interactions. We therefore conclude that protein-protein interactions are attractive in the case of the binary gS-water system. However, since 278 K for the binary gD-water system, the

magnitude of the attraction between gS molecules is 10% smaller than the molecular attraction found for gD. Thus this small reduction of the protein-protein attraction is sufficient to reduce the LLPS temperature below the freezing point of the solution.

The crystallization of thaumatin has been studied extensively both on earth and under microgravity conditions because the protein crystallizes readily. However, there have been no observations of LLPS. We have discovered that by adding small amounts of PEG, we are able to observe LLPS for this model protein. To extend the range of conditions at which LLPS can be observed, we have begun to explore the effect of changing pH. Our preliminary results show that as the pH of the solution is increased, the amount of PEG required to induce LLPS at a given temperature and protein concentration decreases. This observation is consistent with the high value of the isoelectric point of the protein (pI=11). As the pH is increased, the net charge of the protein decreases and there is a corresponding reduction in the electrostatic repulsion between proteins. Therefore, as the pH increased, less PEG is needed to provide the attraction necessary for LLPS to be observed. At pH=10, we approach the isoelectric point, that is the point at which the protein is electrically neutral. It is therefore not surprising that at high pH the protein irreversibly aggregates, even in the absence of PEG. Aggregation close to the isoelectric point is commonly observed in proteins. This investigation showed that observation of LLPS is facilitated close to the isoelectric point. Ref. Annunziata, O., Ogun, O. and Benedek, G. B., "Observation of liquid-liquid phase separation for eye lens gS crystallin," Proc. Natl. Acad. Sci. USA 100, 970-974 (2003).

3. Effect of primary structure modifications on protein crystallization

We have used bovine gB crystallin as a model system to study the role of the individual cysteine residues in the phase separation of the g crystallins. We compared the thermodynamic and kinetic behavior of recombinant wild type protein and the Cys 18 to Ser mutant. We found that the two proteins display essentially identical phase boundaries. The kinetics of crystallization, however, are very different. Despite this difference, the two proteins were found to have the same crystal forms and almost identical x-ray structures. Our results demonstrate that even conservative point mutations can bring about dramatic changes in the kinetics of crystallization. Cysteine residues can lead to protein-protein aggregation by covalent bonds. Thus, this investigation suggests that the replacement of cysteine residues with other conservative amino acid residues can be strategically used to optimize protein crystallization.

Ref. Asherie, N., Pande, J.,Pande, A., Zarutskie, J. A., Lomakin, J., Lomakin, A., Ogun, O., Stern, L. J., King, J. and Benedek, G. B. “Enhanced crystallization of the Cys18 to Ser mutant of bovine gB crystallin” J. Mol. Biol. 314, 663-669 (2001).

Task Progress:

We have also used human bB1-crystallin as a model system to study the role of truncation at the N-terminal peptide on protein phase transitions. This N-terminal can be describes as a flexible arm extended outside the globular compact domains of the protein. We used recombinant DNA techniques to produce native bB1 and truncated bB1DN41. Using circular dichroism, we found that this modification does not change the conformational properties of the remaining protein. One of the results we obtained is that crystallization occurs for truncated bB1DN41 but not for native bB1. This investigation clearly shows that flexible peptide chains significantly interfere with protein crystallization.

Ref. Annunziata, O., Pande, A., Pande, J., Ogun, O., Lubsen, N.H., and Benedek, G.B. “Oligomerization and phase transitions in aqueous solutions of native and truncated human bB1-crystallin,” Biochemistry, 44, 1316-1328 (2005).

4. Effect of oligomerization and self-association on protein phase transitions

We have investigated the phenomena of oligomerization and phase separation in aqueous solutions of native bB1 and its truncated form bB1DN41. These studies have allowed us to compare the phase behavior for bB1 with that of the well-characterized g-crystallin systems, where oligomerization does not occur, and gain insight on general phase behavior of protein aqueous solutions in the presence of reversible oligomerization.

We first used quasielastic light scattering for determining the attraction energy associated with oligomerization. These oligomerization studies showed that the energy of attraction between the bB1DN41 proteins is about 10% greater than that of the bB1’s.

For both bB1 and bB1DN41, we have shown that oligomerization increases as the temperature is lowered. As the temperature is lowered sufficiently, we observe heterogeneous gelation for the bB1DN41 case. We find that heterogeneous gelation differs from a regular liquid-liquid phase transition, such as the one observed in the case of g-crystallins, because it is the result of LLPS accompanied by association and gelation within the protein-rich phase. The gelation process dramatically reduces the rate of liquid-liquid phase separation at concentration higher than about 100mg/ml. Indeed macroscopic phase separation of the two coexisting phases can not be obtained even after centrifugation. At low concentration liquid droplets of concentrated protein neither grow nor coalesce due to gelation. This phase transition is not observed in the case of bB1 aqueous solutions. However, upon the addition of PEG, we observe heterogeneous gelation also for bB1. Our PEG experiments allow us to estimate the difference in phase separation temperatures between bB1 and bB1DN41. The increase in critical temperature from the bB1 case to the bB1DN41 case correlates with the corresponding increase in the magnitude of the oligomerization energy.

We have also investigated the crystallization kinetics for truncated bB1DN41. The crystallization mechanism involves an initial formation of thin rod-like assemblies which then evolve to form crystals. We found that protein solubility with respect to the rod-like assemblies is higher than that taken with respect to crystals. This is consistent with the crystals being more thermodynamically stable. It is important to remark that for concentrations intermediate between these two solubility values crystal nucleation does not occur. We therefore studied the nucleation of the rod-like assemblies by measuring their induction time (inversely proportional to the rate of nucleation). As expected, we find that the induction time decreases as the concentration increases at constant temperature due to the increased supesaturation. However, as the temperature decreases at constant protein concentration, the induction time first decreases and then increases becoming very large in the proximity of the LLPS boundary. The observed minimum in the induction time can be accounted for by considering the oligomerization process and its temperature dependence. These results show that the rate of protein crystallization is significantly affected by the presence of other self-associative processes in solution and the presence of other phase transformation such as LLPS and gelation.

Ref. Annunziata, O., Pande, A., Pande, J., Ogun, O., Lubsen, N.H., and Benedek, G.B. “Oligomerization and phase transitions in aqueous solutions of native and truncated human bB1-crystallin,” Biochemistry, 44, 1316-1328 (2005).

5. Theoretical investigation of nucleation of protein crystals

	The inability to produce high-quality crystals is currently a major hurdle in determining the three-dimensional x-ray structure of proteins. In addition, the crystallization of proteins is involved in diseases such as human genetic cataract. It is therefore important to improve the understanding of the thermodynamics and kinetics of protein crystallization. Some progress has been made through the use of simple models which capture the essential features of proteins interactions. In particular, a globular protein is well described as a hard sphere with short-range, attractive interactions. The phase diagram of such model particles is believed to be generic for globular proteins. There are two possible phase transitions: crystallization and LLPS. LLPS is analogous to the gas-liquid phase transition observed for simple fluids, such as argon. In proteins, however, LLPS is metastable with respect to crystallization, while for simple fluids a stable liquid phase exists. This difference is due to the relatively large size of proteins: if the range of interaction is less than approximately a quarter of the particle diameter, there is no stable liquid phase. Proteins are crystallized from highly supersaturated, but dilute solutions, so as to promote the nucleation rate for crystals while minimizing aggregation. At high supersaturations both liquid and solid phases can nucleate. The question thus arises as to the structure of a nucleus en route to forming a crystal. Operating within the framework of classical nucleation theory, we examine the equilibrium properties of clusters of different sizes. We performed NVT Monte Carlo simulations of hard spheres with short-range interactions using an attractive square-well potential with reduced range l and depth e. For this potential, two particles are said to be in contact if their centers are within the range of interaction. A cluster of size M is then defined as a group of M particles forming a continuous series of contacts. The principal result of our simulations was that for small clusters only liquid-like structures are stable. For a crystalline cluster to form it must be large enough. The transition between a liquid-like and a crystalline cluster results from the competition between the surface and the bulk of the clusters. Our simulations are performed under conditions for which LLPS is metastable i.e. a bulk crystal is more stable than a bulk liquid phase. However, for clusters of finite size, surface effects become important and these favor a liquid-like structure. As the cluster size increases the relative number of particles on the surface decreases, the bulk properties can dominate and the transition from a liquid cluster to a crystalline one occurs. Our findings suggest that a protein crystal should typically nucleate not as a small crystal, but as a more stable disordered aggregate. Only when this aggregate reaches a critical size, can it convert into a crystal. Aggregation per se is not detrimental to the outcome of a crystallization attempt. Instead, the crucial step is the ordering of a critical aggregate which should occur faster than further, unordered growth. Ref. Lomakin, A., Asherie, N. and Benedek, G.B., “Liquid-solid transition in nuclei of protein crystals,” Proc. Natl. Acad. Sci. USA 100, 10254-10257 (2003)
Bibliography Type:	Description: (Last Updated: 04/29/2010)
Articles in Peer-reviewed Journals	Annunziata O, Ogun O, Benedek GB. "Observation of liquid-liquid phase separation for eye lens gammaS-crystallin." Proc Natl Acad Sci U S A. 2003 Feb 4;100(3):970-4. PubMed PMID: 12529503 , Jan-2004
Articles in Peer-reviewed Journals	Lomakin A, Asherie N, Benedek GB. "Liquid-solid transition in nuclei of protein crystals." Proc Natl Acad Sci U S A. 2003 Sep;100(18):10254-7. PubMed PMID: 12925745 , Jan-2004
Articles in Peer-reviewed Journals	Annunziata O, Pande A, Pande J, Ogun O, Lubsen NH, Benedek GB. "Oligomerization and phase transitions in aqueous solutions of native and truncated human beta B1-crystallin." Biochemistry. 2005 Feb 1;44(4):1316-28. PubMed PMID: 15667225 , Feb-2005